Proc. Natl. Acad. Sci. USA
Vol. 78, No. 4, pp. 2427–2431, April 1981
Ecology

Breakage and propagation of the stony coral *Acropora cervicornis*

(reef corals/fragility/wave damage/regrowth)

VERENA TUNNICLIFFE*

Department of Biology, Yale University, New Haven, Connecticut 06520

Communicated by John W. Wells, January 9, 1981

ABSTRACT **Populations of the staghorn coral, *Acropora cervicornis*, often form dense monotypic stands on shallow Caribbean reefs. This coral species has a fragile structure that results in large numbers of broken branches and toppled colonies, especially in high wave activity. Although more than 80% of the corals in the studied population were broken from their bases, most had become reanchored to regrow rapidly. There is little evidence of sexual reproduction, and it appears that this coral has come to dominate much of the Jamaican reef community by propagation through fragmentation.**

The staghorn coral, *Acropora cervicornis*, is a common scleractinian in waters of the Caribbean Sea less than −30 m. In many areas, populations of the coral are extensive enough to form nearly monotypic stands. *A. cervicornis* is fragile and easily broken in high wave activity, and yet it is abundant in areas of relatively high water turbulence. Broken and toppled colonies are found frequently, but this species appears to have the ability to reanchor itself and regrow rapidly. This study was launched to investigate the ability of *A. cervicornis* to survive the impact of wave-induced water flows and the effect of its fragility on its ability to grow, compete, and reproduce.

A. cervicornis has an open, ramose colony form that may reach over a meter in height. It tends to branch, often trichotomously, about once a year and has one of the most rapid of scleractinian growth rates at about 12 cm/yr (ref. 1; unpublished results). Studies of fracture properties have shown that the aragonitic material of acroporid corals is stiff but brittle (2–4). When the material strength is exceeded, fracture is sudden and complete. The drag and inertial forces in wave-generated flows cause the greatest bending and torsional stresses at the small basal attachment points (about 2-cm diameter) of these corals.

This study was conducted on the north coast of Jamaica at Discovery Bay. The bay (see Fig. 1) has a barrier across the north side that is formed by a cresting reef; the bay itself is a large back-reef lagoon. The lobed "haystacks" (5) of the west fore reef support the highest populations of *A. cervicornis* between −5 and −20 m, where interlocking branches can form dense networks. The east fore reef of the bay is a wide platform of low slope, with no major topographic features and a sparse *A. cervicornis* population. Behind the reef crest, where there is low wave activity and high sedimentation rates, the *A. cervicornis* populations are also small. The reef environments and communities of Discovery Bay have been described more completely elsewhere (6–8).

METHODS

In each of the three study areas, a transect (marked in meters) was established perpendicular to the reef crest to traverse the *A. cervicornis* populations (Fig. 1). On the west fore reef and the east back reef, continuous lines 300 m and 130 m in length, respectively, were used. The platform of the east fore reef was too wide for continuous sampling, so a 50-m line was examined at each of five discrete depths for a total of 250 m. Measurements of population densities were made at every 10 m (horizontally) along each of these transects; a 4-m^2 quadrat was used, which incorporated four 1-m^2 replicates. Two measures were made: (*i*) the number of bases of live colonies within each square meter and (*ii*) the total length of live coral branches per square meter.

The *A. cervicornis* colony present under each meter marker of the transects (total, 368 corals) was examined. Height was measured as the maximum distance the coral branches extended above the substratum. Broken surfaces within the colonies were recorded as tip, branch, or basal breaks; the detection of a tip break was somewhat subjective and is defined as the loss of 10 cm or less of the branch. Because these corals regenerate rapidly, evidence of a loss is present for only about 1 year. The third characteristic that was noted for each colony was one of three types of basal attachment: (*i*) unbroken, (*ii*) broken and loose, and (*iii*) broken and subsequently reattached to the substratum.

Some difficulties were encountered in identification of the corals. The juveniles of *A. palmata*, *A. prolifera*, and *A. cervicornis* have similar appearances, but all doubtful cases were scored as *A. cervicornis*. The following year, a few of these corals began to assume the appearance of other species.

RESULTS

Population measurements of *A. cervicornis* are presented in Table 1. There was a high correlation (Pearson $r = 0.80$, $P < 0.001$) between the results of the two methods of population measurement; only the numbers of corals per m^2 are shown here. Maximum measurements of live coral branches on the west fore reef exceeded 10 m of branches per m^2. There, *A. cervicornis* dominated the haystacks between −5 and −20 m. In contrast, it was present on the east fore reef mainly at the −12.5- and −15.3-m sites; the coral community on the bare rock surfaces at less than −10 m was sparse. In the back-reef study area, the *A. cervicornis* population was patchy. In waters deeper than −8 m, it was the only coral found on the sand substratum; it continued down to a depth of −11 m, below which no epibenthic organisms were found.

The height measurements of the corals from the three sites are also shown in Table 1. The mean of heights for the entire group of west fore-reef corals ($\bar{x}$ = 39 cm) differs significantly from both that of the east fore-reef corals ($\bar{x}$ = 27 cm) and that of the back-reef corals ($\bar{x}$ = 27 cm) ($P < 0.001$ for both t tests; measurements were normally distributed). Also noteworthy is the significantly smaller mean size of the corals in the shallowest area of the east fore reef in comparison with the rest of that study area.

The publication costs of this article were defrayed in part by page charge payment. This article must therefore be hereby marked "*advertisement*" in accordance with 18 U. S. C. §1734 solely to indicate this fact.

* Present address: Ocean Ecology, Institute of Ocean Sciences, Sidney, BC, Canada V8L 4B2.

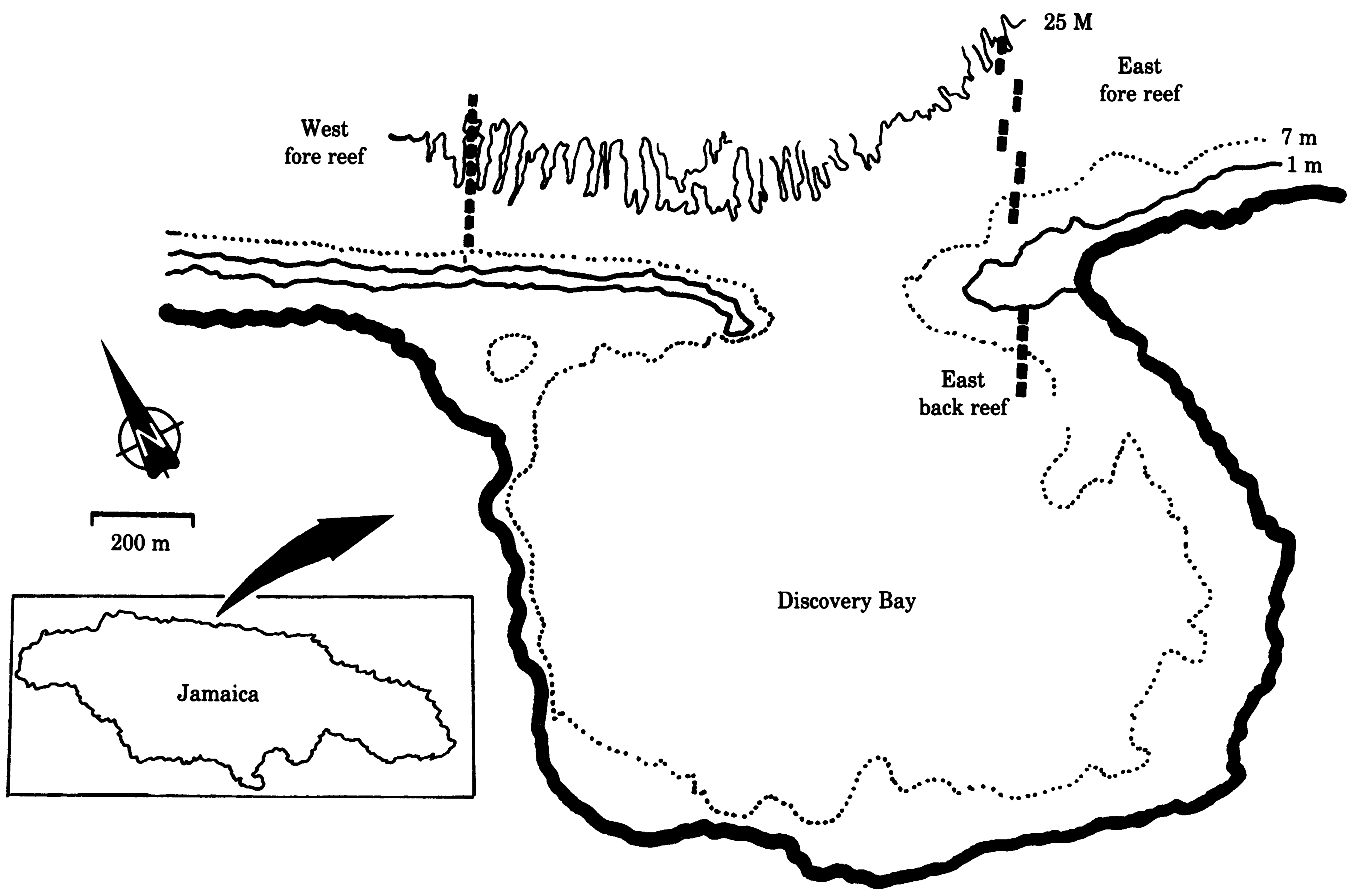

FIG. 1. Discovery Bay, Jamaica. This map shows the locations of the three study sites on the west fore reef, the east fore reef, and the east back reef; the dashed lines indicate the transects. The depth contours are approximate.

Broken Corals. In all of the coral colonies examined, a total of 465 tips and branches were counted as lost. The distribution of the losses among the study areas is illustrated in Fig. 2, where the mean numbers of losses per colony are shown. Branch and tip losses were particularly high on the west fore reef and the deeper back reef. The corals of the intermediate east fore-reef depth showed a moderate amount of damage, whereas those of the shallowest sites showed no damage at all. On the west fore reef, the total number of fractures per colony correlated significantly with colony size (Pearson $r = 0.23$, $P < 0.001$, $n =$ 187) but in the back reef, depth was the only significant variable ($r = 0.57$, $P = 0.001$, $n = 56$).

Table 1. Population densities and mean height of coral colonies at different depths on the Discovery Bay transects

	Population size			Coral height, cm		
Depth, m	No./m^2	SD	N	Mean	SD	N
I. West fore reef						
5.5–10.4	7.1	3.5	32	37.4	14.7	69
10.5–16.4	7.1	3.2	32	41.0	14.6	88
16.5–30.0	4.1	3.8	16	36.7	14.9	30
II. East fore reef						
6.0 and 9.0	0.1	—	Est.	6.4	9.3	12
12.5 and 15.3	2.1	1.9	12	32.0	17.5	79
21.5 and 26.0	0.5	0.7	12	22.8	16.9	37
III. Back reef						
0 – 7.9	0.3	0.7	36	25.9	13.4	29
8.0–11.0	2.0	2.3	16	29.1	11.6	26

The east fore reef was sampled at discrete depths rather than a continuous line. N = number of m^2 quadrats; No. = number of single coral colonies; SD = standard deviation; Est., estimated.

Eighty percent of all of the *A. cervicornis* colonies had been detached at their bases; this condition was particularly common among corals of the west fore reef and back reef, where similar frequencies are seen at all depths.

Reattached Corals. Despite the large number of basal breaks, most colonies were not lying loose on the substratum. Reattachment was achieved by one of two mechanisms: (*i*) live coral tissue overgrew a suitable surface where it touched after the coral fell or (*ii*) encrusting organisms settled on the dead basal areas of the fallen coral and eventually cemented it to the adjacent substratum. These encrusters were almost exclusively coralline algae or a colonial foraminiferan *Gypsina plana*,† both of which precipitate a thin layer of calcium carbonate. A crustose red alga [probably a peyssonneliacean (10)] was present in places of low light incidence.

The histograms for basal breakage in Fig. 2 also show the frequency of colonies that had formed a secondary attachment to the substratum. Eighty percent of the broken corals of the west fore reef had been reattached to the substratum. In the deeper sites of the fore reef, and particularly of the back reef, reattachment of the broken colonies was less common. The type of basal attachment of the *A. cervicornis* colonies differed significantly among the study areas, with the reattached condition

† *Gypsina plana*, first described by Carter (9), forms crusts similar to those of the coralline algae but of a dull grey-brown color. In Discovery Bay, it is common in shallow water and appears to be as important a binder as the corallines in the *cervicornis* zone.

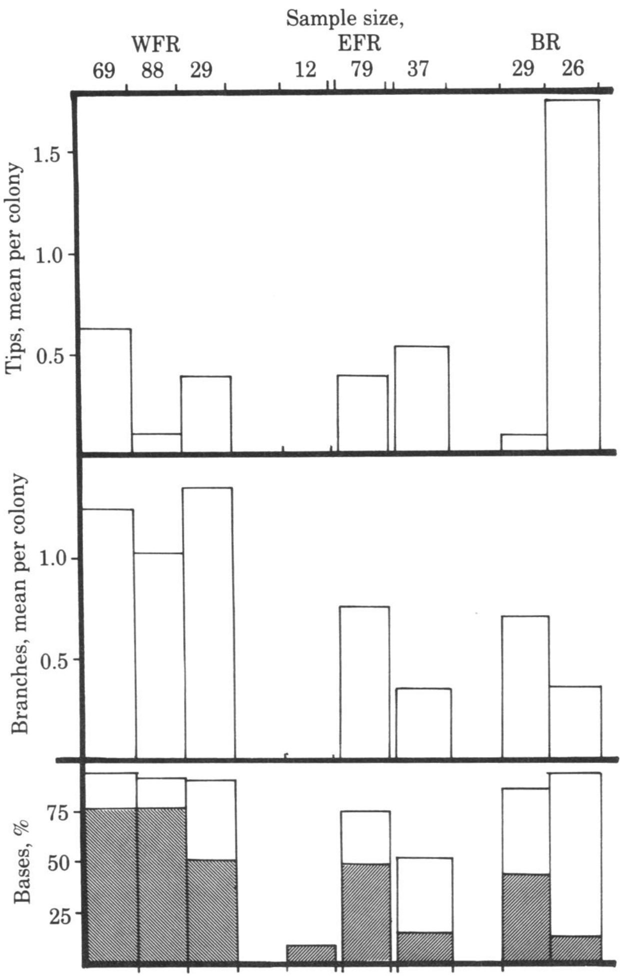

FIG. 2. Frequencies of breaks found in colonies of *A. cervicornis*. The frequencies are shown for the three study areas, west fore reef (WFR), east fore reef (EFR), and back reef (BR). The basal breakage frequencies represent the fraction of the sampled corals found broken, and the shaded area is the fraction that had been reattached to the substratum after breakage. The number of corals examined at each depth is: (*i*) for WFR, 69 at 5.5–10.4 m, 88 at 10.5–16.4 m, and 29 at 16.5–30.3 m; (*ii*) for EFR, 12 at 6.0 and 9.0 m, 79 at 12.5 and 15.3 m, and 37 at 21.5 and 26.0 m; and (*iii*) for BR, 29 at 0–7.9 m and 26 at 8.0–11.0 m.

predominating in the west fore reef, the unbroken condition in the east fore reef, and the broken but unattached condition in the back reef (χ^2, $P < 0.001$).

Small Colonies. *Derived from planulae.* Eighteen *A. cervicornis* colonies less than 10 cm in height were found on the east fore reef and almost all of these were at the shallowest site and of dubious identification. Only one such colony was present on the west fore-reef transect and two in the back reef. A careful search for more juvenile colonies discovered only a few, and they were mostly below −30 m. These corals had undisturbed basal contacts and were the products of recently settled planulae; there were remarkably few.

Derived from fragments. Reattached corals with heights less than the mean are considered to be likely products of fragments or smaller sections of toppled colonies. The height–frequency distributions of the broken corals were examined for information on the success of recruitment by fragmentation; these frequencies are presented in Fig. 3. Inspection of breaks on all colonies revealed 329 broken branches to 136 broken tips. A 3:1 ratio of branches to tips would be expected solely on the basis of their definitions, as there are six size classes of branches but only two

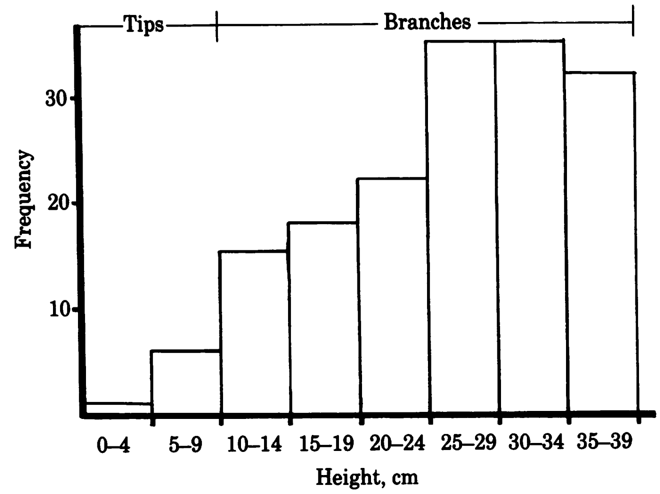

FIG. 3. The height–frequency distribution of the small corals is derived from measurements of 164 corals at all three study areas. Measurements were made at three different times of the year, and there were no obvious age classes. The designations "tips" and "branches" indicate the size classes in which colonies regrowing from these fragments would be found.

of tips (Fig. 3); this assumes random breakage. If all broken fragments survived to regrow, the height frequency distribution would be homogeneous, but there were three significantly different groupings in the sizes 0–9 cm, 10–24 cm, and 25–39 cm ($P < 0.001$ on a χ^2 distribution).

Mortality. Although a total of 465 potential recruits in the form of tips and branches were identified from the scars on the attached colonies, only 164 reattached colonies were discovered in the same areas. Mortality appeared to vary from area to area; the ratio of breaks to reattached colonies on the fore reef was about 4, whereas in the back reef it was 2.

DISCUSSION

Scleractinian corals may reduce the form drag felt in high flow conditions by structural adaptations such as those seen in low, rounded species or the highly oriented palmate colonies (11). In contrast, the structure of *A. cervicornis* has a high bending moment for such a small base and shows a reduced ability to orient its branches (unpublished data). Compounding these problems of stability are the boring sponges that infest the colony bases by excavating tunnels that greatly reduce the ultimate strength of the structure (4). The weakening action of these sponges, coupled with the high wave conditions on the shallower reef, is bound to cause widespread breakage.

The greatest amount of overall damage on the west fore reef is at the shallowest depths, where the wave stresses are the highest. Here, the tallest corals–those with the largest bending moment—sustained the greatest damage. Below −16 m was a 30° slope down which many corals had tumbled and lost branches in the process. In the areas of densest staghorn coral population, broken corals were frequently found lying against neighbors, which also may have toppled or lost a few branches as a result. This "domino" effect accounts for much of the breakage throughout the west fore-reef population. The very small colonies of the shallow east fore reef (that also had few boring sponges) would feel small bending stresses caused by wave forces; breakage among these corals was minimal. Despite their protected position behind the reef crest, the back-reef corals sustained extensive damage, especially the deeper ones. Nearly all corals of the latter group were lodged in the sand; there was little bare substratum available for larval settlement, and it is

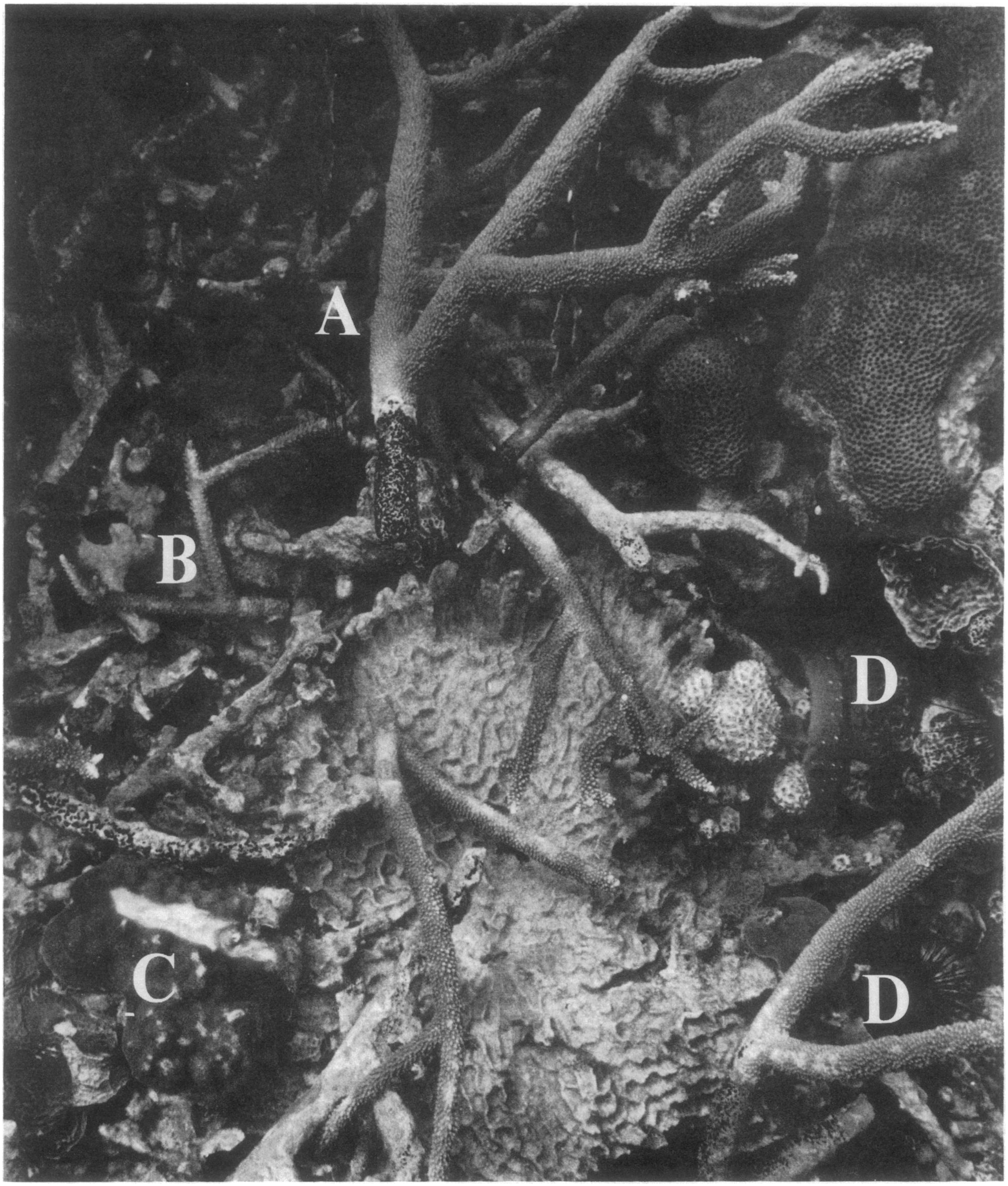

FIG. 4. Broken colonies of *A. cervicornis* at -12 m after a storm. Features: A, a large (about 45 cm) colony broken at a base that is riddled with *Cliona aprica* borings (the black mottling); B, a small fragment not propped off the substratum whose tissues are bleaching; C, another coral, *Porites astreoides*, that has been damaged by a stick of *A. cervicornis*; D, two of the major predators of *A. cervicornis*, a damselfish and the urchin *Diadema antillarum*.

highly probable that these corals had been transported down the lagoonal slope. The deeper the water, the farther the transport and, hence, the greater the amount of damage sustained.

Reattachment of the broken corals was less common in the deepest water. All three reattaching agents, the coral, the algae, and the foraminiferan, have photosynthetic capabilities, so reduced light levels of the back reef and deeper fore reef (unpublished data) likely retard reattachment.

Distinct jumps in numbers of reattached fragments occur at 10 cm and 25 cm. With a growth rate of 12 cm/yr and an annual branching pattern, a branching node is usually found about every 12 cm along the limb. The presence of a branching event within a broken fragment probably increases its chances of survival. If a branch is lying flush against the substratum, it is subject to smothering, predation, or mechanical damage; however, if there is a degree of "three-dimensionality" to the branch (that is, if it is propped up on branch tips and held away from the substratum), less of the broken piece is injured. A 10- to 15-cm-long piece of *A. cervicornis* may include a trichotomous furcation—although usually only dichotomous—and there is more chance that the second branching of the succeeding year at about 25 cm will be in a different plane, thus contributing "props." Such branching patterns may help to explain the variable survivorship among small *A. cervicornis* colonies.

Although toppling usually results in death for most coral species, some corals are able to survive: "The fragment broken off, dropping in a favorable place, would become the germ of another coral plant, its base cementing by means of new coral secretions to the rock on which it might rest ..." (12). Most colonies of *A. cervicornis* of Florida reefs had regenerated from broken pieces of coral and subsequently fused with other colonies (13). Likewise, the Jamaica populations display this ability to regrow and appear to use fragmentation as a method of proliferation. The staghorn structure gives the impression that it is designed to be broken; its success is due to, rather than in spite of, the large number of fractures present.

This vegetative propagation has two major advantages for this coral: (*i*) it allows a rapid establishment of new colonies in a process that is essentially aseasonal and (*ii*) it appears to be an effective method of inhibiting competitors. The presence of other coral species decreased as *A. cervicornis* populations increased (unpublished data) apparently due to shading and wounding from falling branches (Fig. 4). The lower number of *A. cervicornis* colonies on the east fore reef was reflected in the higher diversity of corals and other organisms (S. Ohlhorst, personal communication).

The formation of juvenile *A. cervicornis* colonies from larvae is rare despite the large populations on Caribbean reefs. A recent review of reproduction of corals (14) noted that, of five Pacific shallow-water acroporids, it was the two staghorn species that did not planulate. However, juvenile coral frequencies tend not to reflect the composition of the adult community (15). A few larvally-produced *A. cervicornis* colonies can be found, and it may be that larvae are released under physiological stress (16). The sudden release of embryos at the onset of adverse conditions is common among asexually-reproducing organisms (17–19).

A minimal production of larvae that are readily dispersible means that *A. cervicornis* is not able to avail itself of hospitable but distant habitats. Small barriers can be insurmountable, although the long-distance spread of fragments may be possible (20); it is evident that much of the back-reef *A. cervicornis* population in this study is the product of transport. As a result, the area covered by a clone of this animal may be extensive. On a grander scale, large *A. cervicornis* populations are not present everywhere in the Caribbean. The patchy distribution may be a result of the problem of dispersal; however, the role of the local wave climate is probably of equal importance. *A. cervicornis* is limited by wave activity high enough to promote fragmentation but not so high as to reduce the coral to rubble that cannot regenerate.

The ability of *A. cervicornis* to recover from damage and grow very rapidly may outweigh the disadvantage of nonsexual "reproduction." It is difficult to assess the cost of gamete formation when the requirements are not known, but it may demand an unprofitable allocation of resources so that *A. cervicornis* would not be able to maintain its growth advantage over competing corals. This coral has adapted to its unstable structure and susceptibility to boring sponges, and the resulting vegetative propagation appears to be a major factor in its domination of these Jamaican reefs.

My greatest appreciation for all help and encouragement is due Professors G. E. Hutchinson and W. D. Hartman. Mr. J. Oland gave invaluable support in the field. I also wish to acknowledge the constructive criticisms of P. Tunnicliffe, K. Rylaarsdam, R. Brinkhurst, and N. Knowlton. This work was supported by the Woman's Seamen's Friend Society of Connecticut, Inc., the Canadian Federation of University Women, and National Science Foundation Grant OCE78-06731. This is contribution number 202 from Discovery Bay Marine Laboratory, Jamaica.

1. Shinn, E. (1976) *Environ. Geol.* **4**, 361–364.
2. Vosburgh, F. (1977) *Proc. Third Int. Coral Reef Symp., Miami* **1**, 477–482.
3. Chamberlain, J. A. (1979) *Paleobiology* **4**, 419–435.
4. Tunnicliffe, V. (1979) in *Biologie des Spongiaires*, eds. Lévi, C. & Boury-Esnault, N. (Colloques Internationaux du Centre National de la Recherche Scientifique, Paris), No. 291, 309–315.
5. Kinzie, R. (1973) *Bull. Mar. Sci.* **23**, 93–154.
6. Goreau, T. F. (1959) *Ecology* **40**, 67–90.
7. Goreau, T. F. & Goreau, N. I. (1973) *Bull. Mar. Sci.* **23**, 399–464.
8. Goreau, T. F. & Land, L. S. (1974) in *Reefs in Space and Time*, ed. Laporte, L. F. (Soc. Econ. Pal. Min. Spec. Publ., Tulsa, OK), Vol. 18, pp. 77–89.
9. Carter, H. J. (1978) *Ann. Mag. Nat. Hist. Ser. 4* **20**, 172–176.
10. Adey, W. H. & Vassar, J. M. (1975) *Phycologia* **14**, 55–69.
11. Graus, R. R., Chamberlain, J. A. & Boker, A. M. (1977) *Stud. Geol.* No. 4, 135–153.
12. Dana, J. D. (1890) *Coral and Coral Islands* (Dodd, Mead, New York), 3rd Ed., p. 98.
13. Gilmore, M. D. & Hall, B. R. (1976) *J. Sediment. Petrol.* **46**, 519–522.
14. Stimson, J. S. (1978) *Mar. Biol.* **48**, 173–184.
15. Bak, R. P. M. & van Eys, G. (1975) *Oecologia* **20**, 111–115.
16. Marshall, S. M. & Stephenson, T. A. (1933) *Br. Mus. (Nat. Hist.)* Rep. Great Barrier Reef Exped. **3**, 219–245.
17. Birky, C. W. & Gilbert, J. J. (1971) *Am. Zool.* **11**, 245–266.
18. Mehra, H. R. (1920) *Proc. Zool. Soc. London* **32**, 457–465.
19. Abrahamson, W. G. (1975) *Ecology* **56**, 721–726.
20. Grant, V. & Grant, K. A. (1971) *Evolution* **25**, 144–155.